

LAB 05

Assisted backflip

Backflip: Assisted Launch, PPO Landing,
and Stand Handoff

Environment design · PPO · Honest verification

BUILD → TRAIN → VERIFY → EXPLAIN



What you will build — and what the evidence says

- 01** Use case: Teach controller ownership, transfer learning, PPO diagnostics, and evidence boundaries through a three-stage simulated backflip. The spotter owns launch energy and rotation, the trained PPO actor owns only the landing window, and the pretrained stand policy owns the final hold.
- 02** By the end: describe the scene, map observations to actions, explain the reward, trace one PPO update, run a bounded experiment, and defend an evaluation claim.
- 03** This is not an unassisted learned backflip. The simulator spotter supplies substantial launch energy and rotation.
- 04** Scope: local MuJoCo simulation. Do not deploy these classroom artifacts to a physical robot; hardware transfer requires a separate validated process.

Materials and prerequisites • 1/2

- 01 Microduck MuJoCo model with XML position servos at 50 Hz control
- 02 microduck/policies/alpha_stand.onnx as the exact actor and observation-normalizer source and as the post-landing stand controller
- 03 rlx/rlx/environments/backflip.py for the assisted launch, landing-only reward, stage metrics, and handoff logic
- 04 rlx/rlx/environments/backflip_evaluation.py for the independent physical acceptance gate

Materials and prerequisites · 2/2

- 05 `rlx/rlx/models/backflip_bootstrap.py` for exact teacher transfer, parity checks, reward calibration, and critic fitting
- 06 `rlx/examples/ppo_microduck_studio.py` and `scripts/train-backflip.sh` for training, evaluation, rendering, and export
- 07 `rlx/runs/studio/backflip/backflip-e2e-20260908-v5/` for the retained trained run
- 08 `rlx/artifacts/backflip-e2e-20260908-v5/` for the independent audit, plots, per-seed arrays, and provenance

Environment and policy contract · 1/3

- 01** Simulation only. Protocol spotter-launch-landing-stand-v2 uses explicit MuJoCo generalized external forces during launch; it does not teleport the robot.
- 02** Control contract: float32 observation shape 61 and action shape 14 at 50 Hz. Actions are 14 joint-target offsets in the fixed servo order.
- 03** Observation indices 0:3 are trunk angular velocity; 3:6 projected gravity; 6:20 joint position offsets; 20:34 one-control-step-lagged joint velocity; 34:48 previous raw action; 48:51 twist command; 51:55 head command; 55:61 body command.
- 04** The landing actor receives no extra stage, phase, world position, yaw, launch-force, or contact channel. Composite controller code knows the stage, but the PPO actor still receives the standard 61 values.

Environment and policy contract · 2/3

- 05 Launch duration is 1.2 s. The pitch target ends at 5.3 rad with target rate 5 rad/s; target height rises to 0.32 m and lowers toward 0.22 m before release.
- 06 Launch gains and clamps are pitch k_p 1.5, pitch k_d 0.24, pitch torque limit 1.3 Nm, vertical k_p 90, vertical k_d 8.5, vertical force limit 15 N, lateral damping 3, and roll/yaw damping 0.1.
- 07 Release must reach at least 5.3 rad with backward angular speed between 3.5 and 6.5 rad/s. All external forces and torques are cleared at release.
- 08 Handoff requires at least 5.8 rad credited rotation and 10 consecutive stable PPO samples: both feet supported, no non-foot support, upright projection at least 0.9, trunk height at least 0.105 m, gyro norm below $\sqrt{2}$ rad/s, and no spotter assistance.

Environment and policy contract · 3/3

09

After handoff, `alpha_stand.onnx` must maintain the final 100 consecutive stable samples, equal to 2 s at 50 Hz.

How this case is built: one evidence loop



Failure is data: change one cause, retrain, and repeat the same evaluation gates.

Build the case step by step · 1/3

- 01 Start with the invariant 61-observation/14-action contract and label which signals the landing actor can actually observe.
- 02 Load `alpha_stand.onnx` and verify its 512-256-128 ELU actor structure, 61 inputs, 14 outputs, and observation normalizer before transfer.
- 03 Collect 16 teacher landing episodes of 600 steps each, producing 9,600 calibration samples from assisted-release states.
- 04 Copy the teacher actor and normalizer exactly, require action parity below $1e-4$, and freeze observation normalization.

Build the case step by step · 2/3

- 05 Fit only the fresh critic for 10 epochs with Adam at $3e-4$; keep the transferred actor unchanged during this calibration.
- 06 Reset each training episode through the assisted launch, then expose the release observation to PPO. Assisted preparation steps are outside the PPO transition count and receive zero PPO reward.
- 07 Train only the landing actor. Do not run a stand-policy override during PPO collection.
- 08 Evaluate the complete three-stage controller with deterministic actions and reconstruct the pre-handoff stability streak independently.

Build the case step by step · 3/3

09

Run the landing-only audit with the stand handoff removed so the PPO actor must hold the final 100 samples itself.

10

Compare the final actor against the exact initializer and zero-action control on paired seeds; do not infer learning from showcase success.

11

Render the stage-labeled MP4 and contact sheet, then verify policy, stand-policy, evaluation, and media hashes before using the clip.

PPO: improve the actor without trusting a huge step

$$\text{ratio} = \pi_{\text{new}}(a \mid o) / \pi_{\text{old}}(a \mid o)$$

Maximize $\text{mean}[\min(\text{ratio} \times A, \text{clip}(\text{ratio}, 1 - \epsilon, 1 + \epsilon) \times A)]$

Actor: chooses joint actions. Critic: estimates future discounted reward.

A = advantage. ϵ = clipping range. Old log probabilities stay fixed for each collected batch.

Follow one rollout through the learner

- 01** Collect observations, sampled actions, old log probabilities, values, rewards and episode-boundary flags using the current stochastic actor.
- 02** Compute TD residuals and generalized advantage estimates (GAE). A true terminal has zero continuation value; a timeout may bootstrap the saved final observation.
- 03** Cut advantage traces at reset boundaries. Never bootstrap a timeout from the first observation of the next episode.
- 04** Reuse this batch for PPO minibatch updates, then collect fresh on-policy data. Export the deterministic actor and its observation transform for evaluation.

Paper exercise: count samples and clip the objective

- 01 Illustrative arithmetic, NOT this run: $16 \text{ environments} \times 128 \text{ steps} = 2,048$ fresh transitions per rollout. Four epochs reuse those transitions; they do not create 8,192 new samples.
- 02 With $\epsilon = 0.2$, ratio = 1.4 and $A = +2$: $\min(2.8, 2.4) = 2.4$. The minimized policy term is -2.4 before averaging and additional loss terms.
- 03 With the same ratio and $A = -2$: $\min(-2.8, -2.4) = -2.8$. Explain why clipping is asymmetric once the sign of the advantage changes.
- 04 Run from repo root: `python3 docs/microduck-training-student-book/learning_lab.py --gae`. This is a deterministic arithmetic lab, not new policy training.

Reward design: what the agent is paid to do · 1/2

- 01 name: landing_upright | weight: 3 | sign: reward | definition: $\text{clip}((1 - \text{projected_gravity_z}) / 2, 0, 1)$ | scope: PPO landing stage only
- 02 name: landing_pose | weight: 2 | sign: reward | definition: landing_upright times the mean of equal 0.60-rad and 0.15-rad Gaussian pose kernels | scope: PPO landing stage only
- 03 name: landing_settle | weight: 1 | sign: reward | definition: landing_upright times pose score times $\exp(-\text{sum}(\text{gyro squared}) / 2)$ | scope: PPO landing stage only
- 04 name: landing_joint_speed_penalty | weight: 0.05 | sign: penalty | definition: $-\text{mean}(\min(\text{joint_velocity squared} / 100, 1))$ | scope: PPO landing stage only

Reward design: what the agent is paid to do · 2/2

05

name: landing_action_rate_penalty | weight: 0.02 | sign: penalty | definition: -
mean(min((action - previous_action) squared, 1)) | scope: PPO landing stage only

06

name: landing_action_size_penalty | weight: 0.01 | sign: penalty | definition: -
mean(min(action squared / 16, 1)) | scope: PPO landing stage only

07

name: total_reward | weight: null | sign: bounded sum | definition: 3*landing_upright + 2*landing_pose +
landing_settle + 0.05*landing_joint_speed_penalty + 0.02*landing_action_rate_penalty +
0.01*landing_action_size_penalty; bounded from -0.08 to 6 | scope: Every component is zero during
assisted launch and pretrained stand handoff

Recorded training recipe · 1/2

- 01 config: transition_budget | value: 401408 | detail: 196 complete batches of 16 environments x 128 steps; assisted reset preparation is excluded
- 02 config: ppo_batch | value: 16 environments, 128 steps per environment, 4 minibatches, 2 update epochs
- 03 config: optimizer | value: Adam learning rate 3e-6, maximum gradient norm 0.5, fresh PPO optimizer
- 04 config: discount_and_objective | value: gamma 0.99, GAE lambda 0.95, clip coefficient 0.2, clipped value loss enabled, value coefficient 1.0, normalized advantages enabled

Recorded training recipe · 2/2

- 05 config: exploration | value: initial action standard deviation 0.03, entropy coefficient 0
- 06 config: normalization | value: transferred observation normalization frozen; reward normalization enabled and frozen after teacher calibration; calibrated discounted-return standard deviation 147.2994092878347
- 07 config: episode_and_randomization | value: 12 s episodes, seed 7, XML actuator, dummy vector backend, domain randomization off, observation noise off, action delay off, random yaw off
- 08 config: telemetry_fields | value: phase; env_steps; steps; seconds; optimizer_steps; mean_loss; policy_loss; value_loss; entropy; approximate_kl; clip_fraction; explained_variance; mean_reward; mean_raw_return

Hands-on commands: inspect, smoke, train · 1/2

01

From duck-viewer:

```
npm run dev
```

Start the Studio service before the API reproduction driver.

02

From duck-viewer:

```
node scripts/rlx-dance-api-e2e.mjs --execute --experiment backflip --run backflip-  
reproduction-001 --profile full --recipe-json ../docs/backflip-showcase/recipe.json --  
report ../.restart-lab/backflip-reproduction-001.json
```

Run the documented full Studio train, evaluate, render, and export sequence with a fresh run name.

Hands-on commands: inspect, smoke, train · 2/2

03

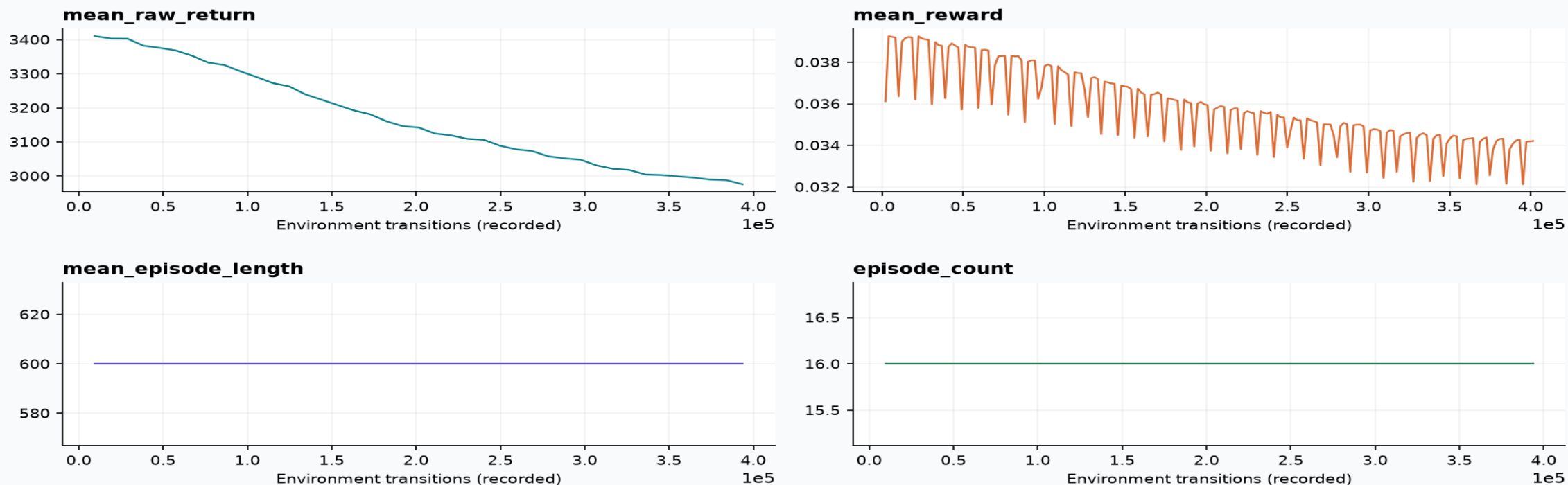
```
From .:  
rlx/.venv-microduck/bin/python rlx/scripts/audit_backflip.py --run  
rlx/runs/studio/backflip/backflip-reproduction-001 --output rlx/artifacts/backflip-  
reproduction-001 --seeds 101,102,103,104,105,106,107,108  
Run the paired showcase, landing-only, initializer, stand, and zero-action audit.
```

04

```
From duck-viewer:  
BACKFLIP_RUN=backflip-e2e-20260908-v5 node scripts/verify-backflip-studio.mjs  
Verify the retained V5 Studio card, guidance, evaluation panel, source-bound video, and  
responsive layout without retraining.
```

Reward: does the logged signal match the task?

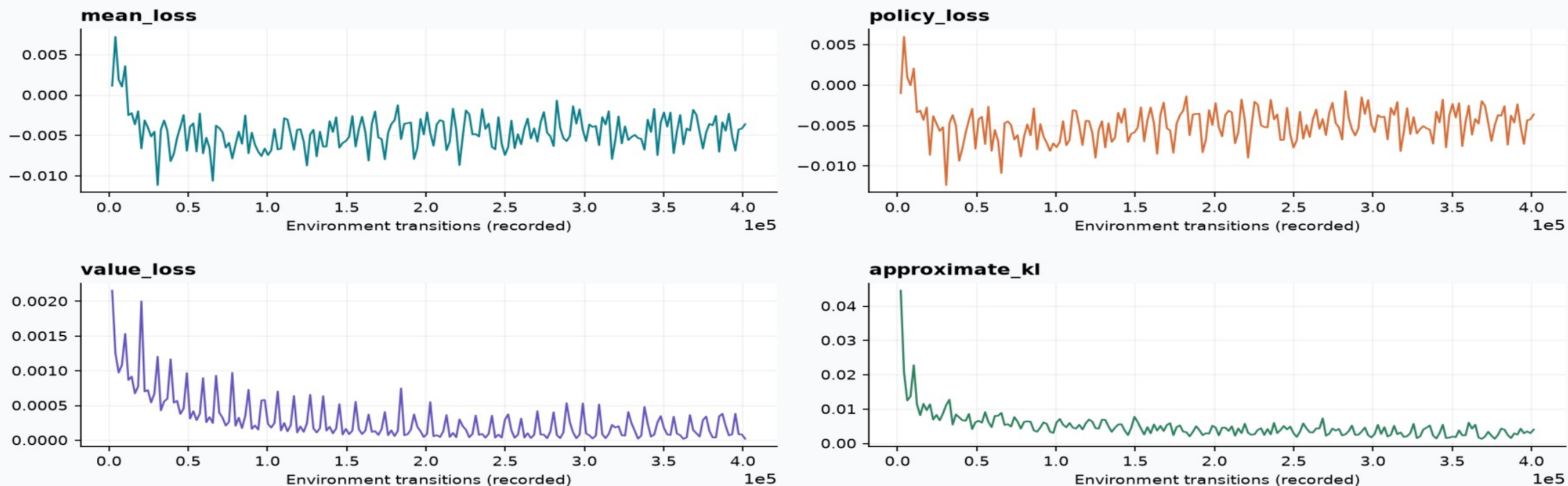
Recorded reward signals — different definitions, not interchangeable



Training reward is an optimization signal, not the acceptance test. See [raw/<case>/](#) for complete recorded fields.

Loss: inspect actor, critic and policy drift

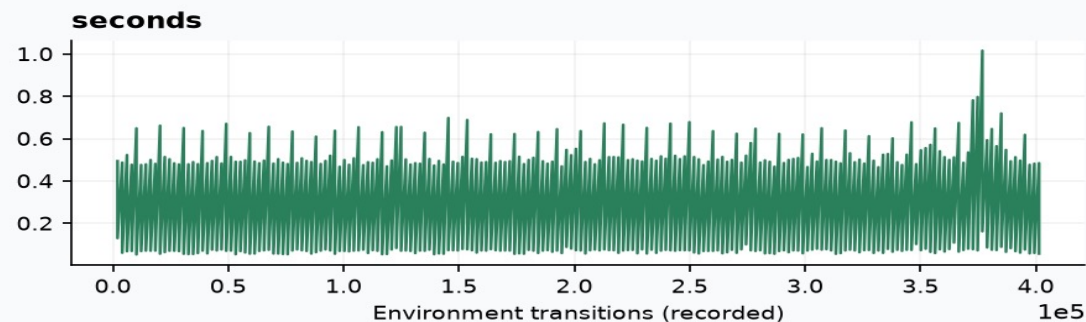
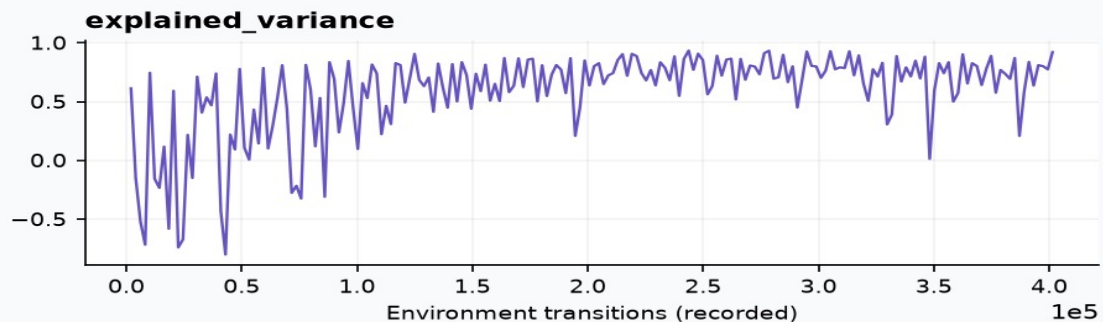
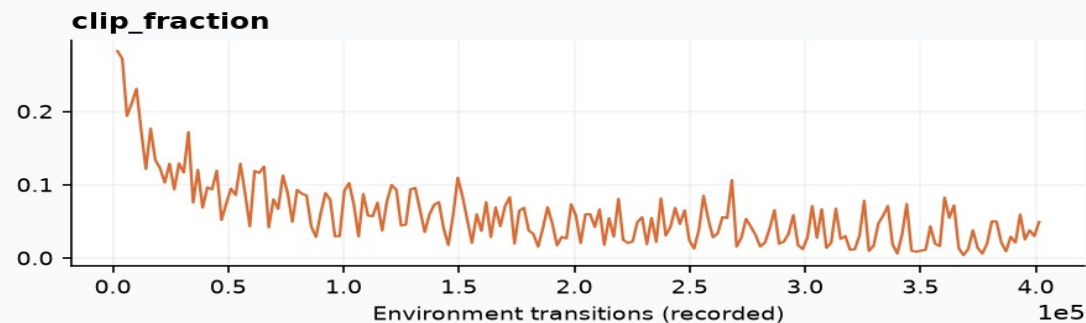
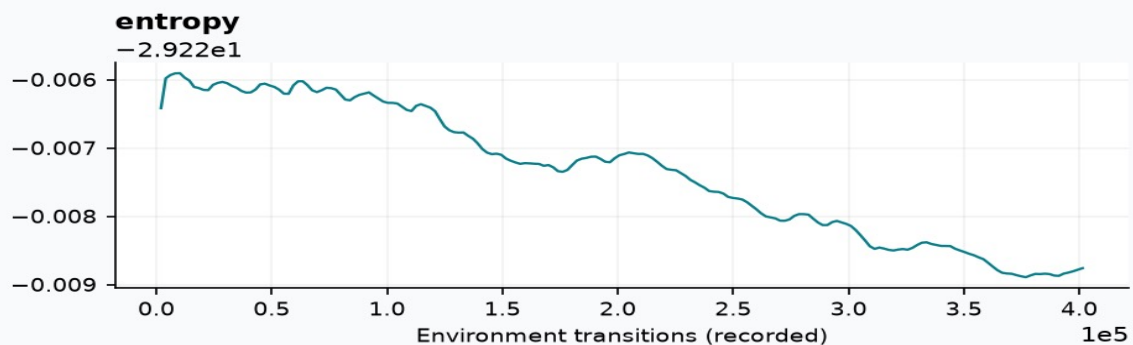
Recorded PPO optimization — raw values, no smoothing



A lower loss does not prove a better robot skill. Loss sign and scale depend on the implementation.

Diagnostics: is the optimizer behaving?

Recorded update diagnostics — absent fields are explicit



Entropy, clipping, explained variance and timing are diagnostics — not a standalone pass/fail certificate.

Metric matrix: numerical audit of this trace · 1/3

- 01 approximate_kl: first 0.04450 → final 0.004050; min 0.001318, max 0.04450; 196 logged samples.
- 02 clip_fraction: first 0.2825 → final 0.04956; min 0.004150, max 0.2825; 196 logged samples.
- 03 entropy: first -29.23 → final -29.23; min -29.23, max -29.23; 196 logged samples.
- 04 explained_variance: first 0.6132 → final 0.9215; min -0.8016, max 0.9343; 196 logged samples.

Metric matrix: numerical audit of this trace · 2/3

- 05 mean_loss: first 0.001169 → final -0.003597; min -0.01114, max 0.007229; 196 logged samples.
- 06 mean_raw_return: first 3411 → final 2976; min 2976, max 3411; 41 logged samples.
- 07 mean_reward: first 0.03612 → final 0.03421; min 0.03213, max 0.03926; 196 logged samples.
- 08 policy_loss: first -0.0009842 → final -0.003618; min -0.01235, max 0.005976; 196 logged samples.

Metric matrix: numerical audit of this trace · 3/3

09

value_loss: first 0.002153 → final 0.00002170; min 0.00001931, max 0.002153; 196 logged samples.

Verification: define success before looking · 1/3

- 01 A smoke profile is only a four-transition pipeline check and is not skill evidence.
- 02 Require deterministic evaluation to report every complete requested episode; reject incomplete, nonfinite, assisted-after-release, unsupported, or failed seeds.
- 03 Check the 600-step three-stage gate separately from the 600-step landing-only gate.
- 04 Confirm `assist_after_release_steps` is zero before crediting PPO landing behavior.

Verification: define success before looking · 2/3

- 05 Confirm handoff rotation is at least 5.8 rad and the pre-handoff PPO stability streak reaches 10 samples.
- 06 Confirm the final 100 landing-only samples meet the physical stability thresholds; a low gyro alone is insufficient.
- 07 Compare final and initializer raw landing returns seed by seed and repeat under fixed action noise before claiming PPO refinement.
- 08 Inspect rollout MP4 and contact sheet for airborne rotation, landing contact, controller stage labels, and final stand hold.

Verification: define success before looking · 3/3

09

Verify ONNX 61-to-14 contract and PyTorch/ONNX action parity, plus policy and stand-policy hash provenance.

Results: what was actually measured • 1/3

- 01 fact: The retained V5 default three-stage evaluation passed 16 of 16 complete 600-step showcase episodes. | source: [rlx/runs/studio/backflip/backflip-e2e-20260908-v5/evaluation.json](#)
- 02 fact: The independent audit passed 8 of 8 showcase episodes on seeds 801-808. | source: [rlx/artifacts/backflip-e2e-20260908-v5/report.md](#)
- 03 fact: With stand handoff removed, the final PPO actor passed 7 of 8 landing-only episodes; seed 808 did not hold all final 100 samples. | source: [rlx/artifacts/backflip-e2e-20260908-v5/report.md](#)
- 04 fact: Paired landing-only mean raw return regressed from 3409.02 for the initializer to 2978.57 for the final actor, a delta of -430.449 or -12.63 percent; 0 of 8 seeds improved. | source: [rlx/artifacts/backflip-e2e-20260908-v5/report.md](#)

Results: what was actually measured · 2/3

- 05** fact: Zero actions passed 0 of 8 landing-only episodes despite a near-zero final gyro; mean final height was about 0.0316 m and upright projection about 0.1351. | source: <rlx/artifacts/backflip-e2e-20260908-v5/report.md>
- 06** fact: Final and initialization ONNX parity maximum absolute action error was $9.53674e-07$, and the 61-input/14-output contract passed. | source: <rlx/artifacts/backflip-e2e-20260908-v5/report.md>
- 07** fact: The first PPO update recorded mean_loss 0.0011691017134580761, policy_loss - 0.0009841816499829292, value_loss 0.002153283334337175, approximate_kl 0.04449799966823775, and clip_fraction 0.282470703125. | source: <rlx/runs/studio/backflip/backflip-e2e-20260908-v5/training-metrics.jsonl>

Results: what was actually measured · 3/3

09 fact: The final PPO update at env_steps 401408 recorded mean_loss - 0.0035966151881439146, policy_loss -0.003618316142819822, value_loss 0.000021701008108720998, approximate_kl 0.004050072617246769, clip_fraction 0.049560546875, and explained_variance 0.9215395247283621. | source: rlx/runs/studio/backflip/backflip-e2e-20260908-v5/training-metrics.jsonl

10 fact: The assisted showcase passed its physical gate but the extended audit verdict is FAIL because PPO refinement and all-seed standalone landing success were false. | source: rlx/artifacts/backflip-e2e-20260908-v5/audit.json

Watch the actual saved rollout



BEFORE PLAY

Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

12.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/backflip-rollout.mp4

Limits and failure analysis • 1/2

- 01 This is not an unassisted learned backflip. The simulator spotter supplies substantial launch energy and rotation.
- 02 The final stand is not owned by PPO; alpha_stand.onnx takes over only after the PPO landing gate.
- 03 Exact teacher transfer means the landing ability was not learned from scratch.
- 04 V5 physical showcase success does not establish PPO improvement; paired landing-only return regressed and one independent landing seed failed.

Limits and failure analysis · 2/2

- 05 The reward can score an upright airborne pose; touchdown and support are enforced by a separate physical gate.
- 06 Randomization, noise, delay, and random yaw were disabled in the retained run.
- 07 The exported backflip.onnx is only the landing-stage actor, not a complete controller and not hardware deployment authorization.
- 08 No result establishes real-robot safety, actuator feasibility, or sim-to-real robustness.

Student lab: change one variable, keep the gates · 1/3

- 01 Draw a three-column timeline and assign every launch, landing, and stand action to spotter, PPO actor, or pretrained stand policy.
- 02 Using the observation table, identify which reward inputs are actor-observable and which acceptance checks are privileged evaluator signals.
- 03 Compute the PPO batch size, number of optimizer steps per update, and number of complete updates in 401,408 transitions.
- 04 Recalculate the reward upper and lower bounds from the six term weights.

Student lab: change one variable, keep the gates · 2/3

- 05 Compare the first and last update fields, then explain why smaller losses or KL do not prove a better landing.
- 06 Use the audit table to calculate the paired percentage change for one seed and compare it with the aggregate -12.63 percent result.
- 07 Inspect ep0.mp4 and list the visual evidence that distinguishes launch assistance, PPO landing, and stand handoff.
- 08 Design a fixed-noise paired evaluation that keeps the physical gate unchanged and tests whether a successor improves the initializer.

Student lab: change one variable, keep the gates · 3/3

09

Write a one-sentence claim that is supported by the evidence and a second sentence that would overclaim.

Submit an experiment, not just a video

- 01** Before training: write one hypothesis, the changed parameter, the fixed baseline, seed list, acceptance criteria and time/transition budget.
- 02** During training: save the exact command, environment versions, initialization, checkpoint and raw reward/loss/KL metrics. Never overwrite the saved reference run.
- 03** After training: evaluate the frozen exported policy on matched seeds; include all failures, aggregate metrics, the full rollout video and a source-bound result table.
- 04** Rubric / 10: reproducibility 2; environment and reward explanation 2; PPO and metrics interpretation 2; honest verification 2; one evidence-based next experiment 2.

Source map and further study · 1/2

01

microduck_local/AGENTS.md

02

docs/student-book/microduck-lab-student-book.md

03

docs/backflip-showcase/REPORT.md

04

docs/backflip-showcase/recipe.json

Source map and further study · 2/2

05

`docs/microduck-training-student-book/chapters/02a-ppo-code-lab.md`

06

`docs/microduck-training-student-book/chapters/02b-gae-workshop.md`

07

`scripts/scenarios/README.md`

08

`README.md`